

Interacting Multipoles of Rank 3 and 4

[illegible]

Simplify deltas:

[illegible]

[illegible]

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned}
& -135135 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot R_\eta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\delta\epsilon\zeta\eta}^A \\
& +10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\delta\epsilon\zeta\zeta}^A \\
& +20790 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\delta\epsilon\epsilon\zeta}^A \\
& +31185 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\delta\delta\epsilon\zeta}^A \\
& +41580 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\gamma\delta\epsilon\zeta}^A \\
& +41580 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\beta\delta\epsilon\zeta}^A \\
& +10395 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{\gamma\delta\epsilon\zeta}^A \\
& +41580 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\delta\epsilon\zeta}^A \\
& +10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{\gamma\delta\epsilon\zeta}^A \\
& +10395 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\alpha\gamma}^B \cdot \Phi_{\beta\delta\epsilon\zeta}^A \\
& -2835 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\delta\delta\epsilon\epsilon}^A \\
& -2835 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\gamma\delta\epsilon\epsilon}^A \\
& -8505 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\gamma\delta\delta\epsilon}^A \\
& -2835 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\beta\delta\epsilon\epsilon}^A \\
& -8505 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\beta\delta\delta\epsilon}^A \\
& -5670 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\epsilon}^A \\
& -11340 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& -2835 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\delta\epsilon\epsilon}^A \\
& -8505 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\delta\delta\epsilon}^A \\
& -5670 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{\gamma\gamma\delta\epsilon}^A \\
& -11340 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\gamma\delta\epsilon}^A \\
& -5670 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\gamma}^B \cdot \Phi_{\beta\beta\delta\epsilon}^A \\
& -11340 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\beta\delta\epsilon}^A \\
& -7560 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& -3780 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{\alpha\gamma\delta\epsilon}^A \\
& +315 \cdot R_z^6 \cdot R_\alpha \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +1260 \cdot R_z^6 \cdot R_\alpha \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +315 \cdot R_z^6 \cdot R_\beta \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +1260 \cdot R_z^6 \cdot R_\beta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\gamma\delta\delta}^A \\
& +315 \cdot R_z^6 \cdot R_\gamma \cdot \Omega_{\alpha\alpha\gamma}^B \cdot \Phi_{\beta\beta\delta\delta}^A \\
& +1260 \cdot R_z^6 \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\beta\delta\delta}^A \\
& +2520 \cdot R_z^6 \cdot R_\delta \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
& +1260 \cdot R_z^6 \cdot R_\delta \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{\alpha\gamma\gamma\delta}^A \\
& +2520 \cdot R_z^6 \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A
\end{aligned} \right] \tag{2d}
\end{aligned}$$

Simplify R:

$$\begin{aligned}
& + \frac{1}{1575} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta} \Omega_{\alpha\beta\gamma}^B \Phi_{\delta\epsilon\zeta\eta}^A = R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned}
& -135135 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzz}^B \cdot \Phi_{zzzz}^A \\
& +10395 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzz}^B \cdot \Phi_{zz\zeta\zeta}^A \\
& +20790 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzz}^B \cdot \Phi_{zz\epsilon\epsilon}^A \\
& +31185 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzz}^B \cdot \Phi_{zz\delta\delta}^A \\
& +41580 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\gamma}^B \cdot \Phi_{zzz\gamma}^A \\
& +41580 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\beta}^B \cdot \Phi_{zzz\beta}^A \\
& +10395 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\beta\beta}^B \cdot \Phi_{zzzz}^A \\
& +41580 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha}^B \cdot \Phi_{zzz\alpha}^A \\
& +10395 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +10395 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& -2835 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzz}^B \cdot \Phi_{\delta\delta\epsilon\epsilon}^A \\
& -2835 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\gamma}^B \cdot \Phi_{z\gamma\epsilon\epsilon}^A \\
& -8505 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\gamma}^B \cdot \Phi_{z\gamma\delta\delta}^A \\
& -2835 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\beta}^B \cdot \Phi_{z\beta\epsilon\epsilon}^A \\
& -8505 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\beta}^B \cdot \Phi_{z\beta\delta\delta}^A \\
& -5670 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\beta\beta}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -11340 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{zz\beta\gamma}^A \\
& -2835 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\epsilon\epsilon}^A \\
& -8505 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\delta\delta}^A \\
& -5670 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -11340 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{zz\alpha\gamma}^A \\
& -5670 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
& -11340 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& -7560 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{zzz\beta}^A \\
& -3780 \cdot R_z^4 \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{zzz\alpha}^A \\
& +315 \cdot R_z^6 \cdot R_z \cdot \Omega_{z\beta\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +1260 \cdot R_z^6 \cdot R_z \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +315 \cdot R_z^6 \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +1260 \cdot R_z^6 \cdot R_z \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\gamma\delta\delta}^A \\
& +315 \cdot R_z^6 \cdot R_z \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{\beta\beta\delta\delta}^A \\
& +1260 \cdot R_z^6 \cdot R_z \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\delta\delta}^A \\
& +2520 \cdot R_z^6 \cdot R_z \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& +1260 \cdot R_z^6 \cdot R_z \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& +2520 \cdot R_z^6 \cdot R_z \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A
\end{aligned} \right] \quad (3a)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned}
& +41580 \cdot R_z^7 \cdot \Omega_{zz\gamma}^B \cdot \Phi_{zzz\gamma}^A + 41580 \cdot R_z^7 \cdot \Omega_{zz\beta}^B \cdot \Phi_{zzz\beta}^A \\
& +41580 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{zzz\alpha}^A - 2835 \cdot R_z^7 \cdot \Omega_{zz\gamma}^B \cdot \Phi_{z\gamma\epsilon\epsilon}^A \\
& -8505 \cdot R_z^7 \cdot \Omega_{zz\gamma}^B \cdot \Phi_{z\gamma\delta\delta}^A - 2835 \cdot R_z^7 \cdot \Omega_{zz\beta}^B \cdot \Phi_{z\beta\epsilon\epsilon}^A \\
& -8505 \cdot R_z^7 \cdot \Omega_{zz\beta}^B \cdot \Phi_{z\beta\delta\delta}^A - 11340 \cdot R_z^7 \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{zz\beta\gamma}^A \\
& -2835 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\epsilon\epsilon}^A \\
& -8505 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\delta\delta}^A - 11340 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{zz\alpha\gamma}^A \\
& -11340 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A - 7560 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{zzz\beta}^A \\
& -3780 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{zzz\alpha}^A + 1260 \cdot R_z^7 \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\gamma\delta\delta}^A \\
& +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\delta\delta}^A \\
& +2520 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& +1260 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A
\end{aligned} \right] \quad (3b)
\end{aligned}$$

$$\xrightarrow{\text{reduced indices}} R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned} &+41580 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{zzz\alpha}^A + 41580 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{zzz\alpha}^A \\ &+41580 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{zzz\alpha}^A - 2835 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A \\ &-8505 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A - 2835 \cdot R_z^7 \cdot \Omega_{zz\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A \\ &-8505 \cdot R_z^7 \cdot \Omega_{zz\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A - 11340 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\ &-2835 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A - 8505 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A \\ &-11340 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A - 11340 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\ &-7560 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{zzz\beta}^A - 3780 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{zzz\alpha}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{\alpha\alpha\beta\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ &+2520 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\ &+2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \end{aligned} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned} &+124740 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{zzz\alpha}^A - 22680 \cdot R_z^7 \cdot \Omega_{zz\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A \\ &-11340 \cdot R_z^7 \cdot \Omega_{z\beta\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A - 34020 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\ &-7560 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{zzz\beta}^A - 3780 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{zzz\alpha}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{\alpha\alpha\beta\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ &+2520 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\ &+2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \end{aligned} \right] \quad (3d)$$

Simplify Oktopoles:

$$+\frac{1}{1575} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta} \Omega_{\alpha\beta\gamma}^B \Phi_{\delta\epsilon\zeta\eta}^A = R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned} &+124740 \cdot R_z^7 \cdot \Omega_{yzz}^B \cdot \Phi_{yzzz}^A - 22680 \cdot R_z^7 \cdot \Omega_{yzz}^B \cdot \Phi_{yz\beta\beta}^A \\ &-11340 \cdot R_z^7 \cdot \Omega_{yzz}^B \cdot \Phi_{yz\alpha\alpha}^A - 34020 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\ &-7560 \cdot R_z^7 \cdot \Omega_{y\alpha\alpha}^B \cdot \Phi_{yzzz}^A - 3780 \cdot R_z^7 \cdot \Omega_{y\beta\beta}^B \cdot \Phi_{yzzz}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{\alpha\alpha\beta\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ &+1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ &+2520 \cdot R_z^7 \cdot \Omega_{y\alpha\alpha}^B \cdot \Phi_{yzz\gamma\gamma}^A + 1260 \cdot R_z^7 \cdot \Omega_{y\beta\beta}^B \cdot \Phi_{yzz\gamma\gamma}^A \\ &+2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \end{aligned} \right] \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{array}{l} -34020 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A + 1260 \cdot R_z^7 \cdot \Omega_{z\beta\gamma}^B \cdot \Phi_{\alpha\alpha\beta\gamma}^A \\ + 1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ + 1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ + 2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \end{array} \right] \quad (4b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned} &-34020 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A + 1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ &+ 1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ &+ 1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \end{aligned} \right] \quad (4c)$$

$$\xrightarrow{\text{added up}} R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} -34020 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\ +2520 \cdot R_z^7 \cdot \Omega_{z\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \end{bmatrix} \quad (4d)$$

Simplify Hexadecapoles:

$$+ \frac{1}{1575} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta} \Omega_{\alpha\beta\gamma}^B \Phi_{\delta\epsilon\zeta\eta}^A = R_z^{-15} \cdot \frac{1}{1575} \cdot \begin{bmatrix} -34020 \cdot R_z^7 \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\ +2520 \cdot R_z^7 \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \\ +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\ +2520 \cdot R_z^7 \cdot \Omega_{z\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A \end{bmatrix} \quad (5a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-15} \cdot \frac{1}{1575} \cdot [+2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\gamma}^B \cdot \Phi_{z\alpha\beta\gamma}^A] \quad (5b)$$

Multiply Out Sum:

$$+ \frac{1}{1575} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta} \Omega_{\alpha\beta\gamma}^B \Phi_{\delta\epsilon\zeta\eta}^A = R_z^{-15} \cdot \frac{1}{1575} \cdot \left[\begin{aligned} &+2520 \cdot R_z^7 \cdot \Omega_{xxx}^B \cdot \Phi_{xxxz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xxy}^B \cdot \Phi_{xxyz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xxz}^B \cdot \Phi_{xxzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xxy}^B \cdot \Phi_{xxyz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xyy}^B \cdot \Phi_{xyyz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xyz}^B \cdot \Phi_{xyzx}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xxz}^B \cdot \Phi_{xxzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xxy}^B \cdot \Phi_{xxyz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xzz}^B \cdot \Phi_{xzzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xxy}^B \cdot \Phi_{xxyz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xyy}^B \cdot \Phi_{xyyz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xyz}^B \cdot \Phi_{xyzx}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xyy}^B \cdot \Phi_{xyyz}^A + 2520 \cdot R_z^7 \cdot \Omega_{yyy}^B \cdot \Phi_{yyyz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{yyz}^B \cdot \Phi_{yyzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xyz}^B \cdot \Phi_{xyzx}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{yyz}^B \cdot \Phi_{yyzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{yzz}^B \cdot \Phi_{yzzz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xxz}^B \cdot \Phi_{xxzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xyz}^B \cdot \Phi_{xyzx}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xzz}^B \cdot \Phi_{xzzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{xyz}^B \cdot \Phi_{xyzx}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{yyz}^B \cdot \Phi_{yyzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{yzz}^B \cdot \Phi_{yzzz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{xzz}^B \cdot \Phi_{xzzz}^A + 2520 \cdot R_z^7 \cdot \Omega_{yzz}^B \cdot \Phi_{yzzz}^A \\ &+ 2520 \cdot R_z^7 \cdot \Omega_{zzz}^B \cdot \Phi_{zzzz}^A \end{aligned} \right] \quad (6a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-15} \cdot \frac{1}{1575} \cdot [+0] \quad (6b)$$

Combining everything:

$$+\frac{1}{1575} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta\eta} \Omega_{\alpha\beta\gamma}^B \Phi_{\delta\epsilon\zeta\eta}^A = +0 \quad (7)$$